

5	(a)(i)	$R_R = 105\Omega$ $R_{Th} = 50\ \Omega$	A1 A1
	(a)(ii)	$V = \frac{50}{105+50} \times 12$ $= 3.87\text{ V}$	C1 A1
	(a)(iii)	The p.d across resistor and thermistor is equal so this implies that the resistance across each component is equal. Hence the temperature is $50\text{ }^\circ\text{C}$.	C1 A1
	(b)(i)	p.d across length $x = \frac{x}{l} E$ Since the appliance is parallel to length of x , p.d is the same. Thus p.d across the appliance is $x = \frac{x}{l} E$	M1 A1
	(b)(ii)	p.d across appliance = $20 / 100 \times 5.0 = 1.0\text{ V}$ current = $1/10.0 = 0.10\text{ A}$	A1
	(b)(iii)	$\varepsilon = 45.0 / 100 \times 5.0 = 2.25\text{ V}$	A1

6	(a)	A magnetic field is a region of space in which a magnetic force is experienced	A1
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